

Reducing Misreporting of Health Information through Novel Positive Appeals

Extended Abstract

Context

Misreporting of health behaviors — including diet — poses serious risks, as healthcare professionals may make flawed clinical judgments based on inaccurate data (Ritter et al., 2001). Additionally, inaccurate self-reported data also poses validity threats to studies on information systems (de Reuver & Bouwman, 2015). Misreporting is poised to become far more common and consequential as patients increasingly rely on mHealth technologies to self-report their health data to physicians (Portz et al., 2024). In the field of Information Systems, more proactive approaches, such as fear appeals — messages designed to discourage misreporting by highlighting negative consequences — have been proposed to discourage fraudulent behavior (Wall & Buche, 2017). However, in sensitive domains such as diet or mental health, fear appeals risk inducing fear or feelings of being demeaned, which may undermine honest reporting rather than encourage it (Hastings et al., 2004).

Gap

Existing approaches to encouraging honest self-reporting in the information systems literature exhibit two characteristics that motivate the present study: a reliance on threat-based rather than positive framing, and limited direct attention to self-reported health data as an outcome.

Reliance on Threat-Based Framing

The dominant intervention design for discouraging dishonest behavior, rooted in Protection Motivation Theory (PMT) and supported by an extensive body of 383 studies (Lowry et al., 2023; Wall & Buche, 2017), relies on fear appeals. Yet threat-based framings can be counterproductive in sensitive health domains, provoking defensiveness rather than compliance (Hastings et al., 2004) and showing limited empirical effectiveness in deterring health misreporting (Yu, 2023).

Limited Attention to Self-Reporting

Beyond the framing limitation, honest self-reporting has received comparatively little direct attention as an intervention target. PMT research, including meta-analyses (Lowry et al., 2023) and empirical studies (Boss et al., 2015), has advanced our understanding of security policy compliance, but focuses on behavioral compliance rather than truthful disclosure. Keller (2006) similarly demonstrates the value of motivating health behaviors such as screening and preventive action — a valuable foundation, though one developed for behavioral initiation rather than honest disclosure. Together, these literatures leave encouragement-based approaches to honest self-reporting as an underexplored alternative.

Literature	Positive Framing	Actionable Coping Guidance	Targets Honest Self-Reporting
Lowry et al., 2023; Wall & Buche, 2017	x	v	x
Keller 2006	v	v	x
Our research	v	v	v

Table 1. Research Gap

Proposed Approach

To address the aforementioned gaps, this study proposes an Encouragement-Offering framing that extends Fear Appeal Theory in health contexts by combining its actionable coping strategies and response cost considerations with the aspiration mindset from Regulatory Focus Theory.

Regulatory Focus Theory

Drawing on Regulatory Focus Theory (Higgins, 1997), we replace the threat appraisal central to conventional fear-appeal designs with a promotion-focused orientation, reorienting respondents toward the personal gains of honest disclosure and avoiding the defensive backfires that threat-based framings often provoke.

Fear Appeal Theory

We retain and repurpose the coping appraisal components of Fear Appeal Theory (Wall & Buche, 2017; Keller, 2006), preserving self-efficacy messages that reassure respondents it is easy to report honestly and response-efficacy messages that show them how to do so. We further incorporate response cost as a structural safeguard by administering self-report measures through Response Activation Model (RAM) slider bars (Weinmann et al., 2022; Yu, 2023), which introduce deliberate lags as the slider is dragged — giving respondents time to reflect and raising the cost of casual adjustments.

To our knowledge, no prior work has combined Regulatory Focus Theory's promotion orientation with Fear Appeal Theory's coping appraisal in a single intervention targeting honest self-reporting. If effective, this approach would offer an empirically grounded alternative to threat-based frameworks and extend the scarce literature on honest self-reporting in sensitive domains.

Theory	Constructs	Implementation Implication	Citation
Self-Regulatory Theory	Promotion-focus orientation	Replacing threat appraisal from Fear Appeal with positive constructs to prevent backfires	Higgins, 1997
Fear Appeal Theory	Self-efficacy	Inform “it is easy to report honestly”	Wall & Buche, 2017; Keller 2006
	Response-efficacy	Inform “how to report honestly”	Wall & Buche, 2017; Keller 2006
	Response Cost	Raised response costs of moving casually	Weinmann et al., 2022; Yu, 2023
Table 2. Encouragement Offering Components			

Method

To test the causal effect of the Encouragement-Offering approach, we plan to recruit 400-600 respondents and will randomly assign respondents to a Control Group or an Encouragement-Offering Group, with control participants completing the survey without the promotion-focused induction, coping messages, or RAM slider bar. After completing the diet self-report, all respondents will be shown their health score — the sum of their Likert scale responses — and asked to report it to a medical institute on a separate platform. Because respondents are reporting to a clinical audience, impression management concerns are likely to motivate some to inflate their score (Leary & Kowalski, 1990), creating the conditions for misreporting. Score mismatches between the score shown on the survey and the platform will be considered as misreports. Platform reminders and a reference link to the survey score will rule out typos and memory errors. A significantly lower misreporting rate in the Encouragement-Offering group would support the effectiveness of the approach.

We are further interested in how dishonest and honest respondents differ in mouse movement behavior. We will record pixel-level mouse movements and event types, and analyze them using established metrics including Reaction Time (the interval between question presentation and response click; Mazza et al., 2020), Movement Deviation (the divergence of the actual cursor path from the shortest path; Weinmann et al., 2022), and additional features such as acceleration and trajectory shape (Jenkins et al., 2025), examining whether dishonest respondents exhibit distinct behavioral signatures.

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